

Complete AI Fitness, Nutrition & Lifestyle Coach Knowledge Base

Purpose: A retrieval-friendly knowledge base for an AI chatbot that can analyze food photos, answer nutrition questions, explain gym exercises, build safe general training structures, discuss meal planning, support healthy habits, and provide general fitness/lifestyle coaching.

Important: This is educational information, not medical diagnosis or treatment. Photo-based calorie and macro estimates are approximate. The chatbot must not encourage dangerous dieting, fasting, purging, excessive exercise, body shaming, or weight-loss behaviors in minors. For medical conditions, serious symptoms, significant injuries, eating concerns, allergies, or individualized therapeutic nutrition, refer to a qualified healthcare professional.

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1. AI Coach Identity & Core Rules

Role

You are a supportive nutrition, fitness, exercise-education, and lifestyle assistant. Use the knowledge base as the primary source for general guidance. Be practical, respectful, and non-judgmental.

No false precision

Never present estimated calories from a photo as exact. Use ranges or approximate values when portion size, ingredients, or cooking method are uncertain.

No diagnosis

Do not diagnose diseases, injuries, eating disorders, nutritional deficiencies, hormonal disorders, or other medical conditions.

No prescriptions

Do not prescribe medication, supplements, therapeutic diets, or medical treatment.

Healthy language

Avoid shame, guilt, fear, body comparisons, appearance pressure, or labeling foods as morally good or bad. Focus on nourishment, function, enjoyment, consistency, and health.

Unknown information

If the knowledge base does not contain enough information, say that the answer cannot be determined reliably rather than inventing facts.

2. Food Photo Analysis & Calorie Estimation

Workflow

1) Identify visible foods. 2) Separate meal components. 3) Estimate portion size. 4) Infer likely cooking method. 5) Identify visible oils, sauces, toppings, and drinks. 6) Estimate calories and macros. 7) Assign confidence. 8) Explain uncertainty. 9) Ask a follow-up question when additional information would substantially improve accuracy.

Photo response format

Detected foods: list items. **Estimated portions:** grams, pieces, cups, or a serving estimate. **Estimated calories:** range when uncertain. **Macros:** approximate protein, carbohydrates, and fat when possible. **Confidence:** low, medium, or high. **Missing information:** identify what could change the estimate.

Hidden calories

Photos may not show cooking oil, butter, sugar, mayonnaise, dressings, sauces, marinades, cheese, sweetened drinks, or ingredients inside mixed dishes. These can materially change an estimate.

Portion estimation

Use visible size comparisons and common serving references cautiously. Do not claim exact grams from a photo unless a known scale or labeled serving is provided.

Mixed dishes

For pizza, sandwiches, pasta dishes, curries, casseroles, rice bowls, desserts, and similar meals, break the meal into plausible components and explain that recipes vary.

Confidence

High confidence means the food and portion are clear and simple. Medium means the food is identifiable but portion or preparation is uncertain. Low means important components are hidden or ambiguous.

Recommended disclaimer

This is an estimate based on what is visible in the photo. Exact calories depend on portion size, ingredients, cooking method, oils, sauces, toppings, and recipe.

3. Calories, Energy & Nutrition Labels

Calories

Food energy is commonly expressed in kilocalories (kcal). Calories describe energy, not food quality by themselves.

Energy needs

Energy requirements vary with age, growth, body size, activity, sleep, health, environment, and individual biology.

Nutrition labels

Use serving size, servings per container, calories, protein, carbohydrate, fat, fiber, sodium, and ingredient information to understand packaged foods. Compare products using the same serving basis.

Serving vs portion

A serving is a standardized amount listed by a manufacturer or reference. A portion is the amount actually eaten. They are not always the same.

Calorie counting limitation

Calories are only one part of nutrition. Food variety, protein, fiber, micronutrients, hydration, and eating patterns also matter.

4. Macronutrients & Fiber

Protein

Protein supports growth, tissue maintenance and repair, enzymes, hormones, immune function, and many body processes. Sources include eggs, dairy, fish, poultry, meat, beans, lentils, peas, tofu, nuts, and seeds.

Carbohydrates

Carbohydrates are an important energy source. Sources include grains, bread, rice, pasta, potatoes, fruit, beans, and vegetables. Fiber-rich carbohydrate foods can support digestive health.

Dietary fat

Fat is essential for normal body functions and helps absorb fat-soluble vitamins. Sources include nuts, seeds, olive oil, avocado, fish, eggs, and dairy. Avoid describing all fats as unhealthy.

Fiber

Fiber is found in vegetables, fruit, whole grains, legumes, nuts, and seeds. It supports normal digestion and can contribute to fullness.

Balanced intake

A varied eating pattern can provide all three macronutrients. Exact needs differ among people and situations.

5. Micronutrients, Hydration & Supplements

Vitamins and minerals

Important nutrients include iron, calcium, vitamin D, vitamin C, folate, potassium, magnesium, zinc, and vitamin B12. Requirements vary by age and individual circumstances.

Hydration

Water is the primary hydration choice for most people. Needs increase with heat, sweating, and physical activity. Use thirst and activity/environment cues as practical guides.

Electrolytes

Sodium and other electrolytes are involved in fluid balance and normal body function. Most people can obtain electrolytes through food; special circumstances such as prolonged intense activity may require individualized advice.

Supplements

Supplements are not automatically necessary. Some can interact with medicines or be harmful at high doses. For suspected deficiency or medical needs, seek professional assessment.

Protein powders

Protein powder is a convenient food product for some people, but it is not required for fitness. Whole-food protein sources can also meet needs.

6. Food Groups & Balanced Meals

Balanced plate

A practical general structure is vegetables and/or fruit, a protein-rich food, a carbohydrate source such as grains or potatoes, and a source of fats as appropriate.

Vegetables and fruit

Choose variety in types and colors. Fresh, frozen, and appropriately prepared foods can all contribute to a balanced diet.

Whole grains

Examples include oats, whole-grain bread, brown rice, bulgur, barley, and whole-grain pasta.

Protein foods

Examples include eggs, fish, poultry, lean meats, milk, yogurt, cheese, beans, lentils, tofu, nuts, and seeds.

Treat foods

Higher-sugar, higher-salt, or higher-fat foods can fit into an overall balanced pattern. Avoid guilt-based messaging.

7. Diet Patterns & Meal Planning

Mediterranean-style

Emphasizes vegetables, fruits, whole grains, legumes, nuts, seeds, olive oil, and fish, with moderate amounts of dairy and poultry.

Vegetarian

Avoids meat and may include eggs and dairy depending on the individual. Variety is important for protein, iron, calcium, vitamin B12, zinc, and omega-3 sources.

Vegan

Avoids animal-derived foods. A well-planned pattern requires attention to vitamin B12 and other nutrients; fortified foods or professional guidance may be appropriate.

Low-carbohydrate

Restricts carbohydrates to varying degrees. It is not automatically healthier or suitable for everyone.

Keto

A very low-carbohydrate pattern. It can be difficult to implement and is not appropriate for everyone. Do not prescribe it to minors.

Intermittent fasting

Uses scheduled eating windows. It may be unsuitable for children and adolescents, people with certain medical conditions, or people with eating concerns. The chatbot should not encourage minors to fast for weight control.

Meal planning principles

Plan meals around variety, protein foods, vegetables/fruit, carbohydrate sources, healthy fats, hydration, and personal preferences. Use flexible templates rather than rigid rules.

Meal prep

Prepare convenient staples such as cooked grains, washed vegetables, beans, eggs, or appropriately stored protein foods. Follow food-safety guidance.

8. General Meal-Plan Templates

Template A — Balanced day

Breakfast: oats or whole-grain cereal + milk/yogurt + fruit. Lunch: protein food + rice/potato/grain + vegetables. Snack: fruit + yogurt or nuts. Dinner: eggs/fish/chicken/beans + whole-grain or potato + vegetables.

Template B — Vegetarian

Breakfast: oats + yogurt or fortified plant milk + fruit. Lunch: lentils/beans/tofu + grain + vegetables. Snack: fruit + nuts. Dinner: eggs/dairy or plant protein + vegetables + grain/potato.

Template C — Active day

Include regular balanced meals and, if useful, a carbohydrate-containing snack before activity and a balanced meal or snack afterward. Exact timing depends on the activity and individual tolerance.

Important

These are educational templates, not individualized calorie-restricted plans. For children and adolescents, prioritize growth, nourishment, activity, sleep, and professional guidance when individualized planning is needed.

9. Pre-Workout & Post-Workout Nutrition

Before training

A meal or snack containing carbohydrate and, when appropriate, protein can provide energy. Choose foods that are comfortable to digest.

Timing

There is no single perfect timing for everyone. A larger meal may be eaten earlier and a smaller snack closer to exercise.

After training

Rehydrate and eat a balanced meal or snack when hungry. Include protein-rich and carbohydrate-containing foods as part of the overall diet.

Hydration during training

Drink according to thirst and conditions. Longer or intense sessions, heat, and heavy sweating can increase fluid needs.

10. Gym Fundamentals

Fitness components

Fitness can include aerobic endurance, muscular strength, muscular endurance, mobility, flexibility, balance, coordination, and general physical activity.

Warm-up

Use several minutes of easy movement and activity-specific preparation. Gradually increase intensity.

Cool-down

Gradually reduce intensity. Gentle mobility may feel comfortable; stretching should not be forced.

Progressive overload

Gradually increase a training demand through resistance, repetitions, duration, control, range of motion, or technical quality.

Technique first

Use controlled movement, appropriate resistance, and a manageable range of motion. Good technique is more important than lifting a heavier load.

Rest between sets

Rest enough to perform the next set with good technique. Required rest varies by exercise, goal, fitness level, and intensity.

Training consistency

Consistent manageable training is generally more useful than occasional extreme workouts.

11. Exercise Categories & Movement Patterns

Squat pattern

Examples: bodyweight squat, goblet squat, supported squat, and appropriate machine squat.

Hinge pattern

Examples: hip hinge, light Romanian deadlift pattern, and glute bridge. Technique should be learned before adding substantial load.

Push pattern

Examples: wall push-up, incline push-up, push-up, chest press, and shoulder press using appropriate resistance.

Pull pattern

Examples: resistance-band row, cable row, machine row, and assisted pull-up.

Carry pattern

Examples: light carries using safe, manageable loads and good posture.

Core/bracing

Examples: plank variations, dead bug, bird dog, and controlled anti-rotation exercises.

Lower body

Examples: step-ups, lunges, squats, bridges, and calf raises.

Upper body

Examples: rows, presses, push-ups, pulldowns, and light accessory exercises.

Cardio

Walking, cycling, swimming, jogging, dancing, rowing, and sports can develop cardiovascular fitness.

Mobility

Controlled joint movements through comfortable ranges can support movement quality.

12. Training Styles

Full-body

Train major movement patterns within each session. Useful when training only a few days per week.

Upper/Lower

Alternate upper-body and lower-body sessions. Useful for people who can train multiple days per week.

Push/Pull/Legs

Organizes sessions by movement emphasis. Training frequency should match recovery and experience.

Circuit training

Perform several exercises in sequence with controlled rest. Useful for combining strength and conditioning.

Strength-focused

Prioritize technique, controlled heavy-enough resistance, adequate rest, and gradual progression. Avoid maximal attempts without qualified supervision.

Hypertrophy-focused

Uses repeated resistance-training sets with sufficient recovery and progressive challenge. Do not promise a particular body shape.

Endurance-focused

Uses manageable resistance and/or longer activity duration to improve the ability to sustain effort.

13. Beginner Gym Structure

Beginner principles

Start with manageable sessions, learn movement patterns, avoid excessive volume, and allow recovery.

Example structure

A beginner session can include: warm-up → squat/leg movement → push movement → pull movement → hinge/bridge → core → optional easy cardio → cool-down.

Progression

When an exercise becomes comfortable with good technique, increase repetitions, resistance, control, or range gradually.

Supervision

Beginners should ask a qualified trainer for help learning unfamiliar equipment, especially heavy free-weight exercises.

14. Intermediate & Advanced Training Concepts

Intermediate

Progress by managing training volume, exercise selection, intensity, recovery, and consistency. Keep technique reliable.

Advanced

More complex programming can use planned variation, training blocks, and carefully managed volume/intensity. Advanced methods should be used with appropriate experience and coaching.

Deload

A planned reduction in training demand can be used when fatigue accumulates. The exact method depends on the training program.

Exercise substitution

Replace an exercise with another that trains a similar movement pattern when equipment, comfort, skill, or accessibility requires it.

15. Training Frequency, Sets, Reps & Rest

Frequency

Training frequency depends on goals, age, experience, schedule, and recovery. More sessions are not automatically better.

Sets and repetitions

There is no single rep range that is required for all goals. Use repetitions that can be performed with controlled technique and an appropriate challenge.

Rest

Shorter rests may increase conditioning demands; longer rests can help maintain performance on demanding strength sets.

Effort

Avoid repeatedly training to complete failure. Especially for beginners, stopping with controlled technique and some reserve is often appropriate.

Progress tracking

Track exercises, repetitions, resistance when relevant, technique, energy, and how the session felt.

16. Cardio Training

Steady-state cardio

Continuous moderate activity such as walking, cycling, swimming, or jogging.

Intervals

Alternating periods of harder and easier activity. Intensity should be appropriate to the person's experience and health.

Low-impact cardio

Walking, cycling, swimming, elliptical machines, and similar activities can reduce impact while providing cardiovascular exercise.

Sports

Football, basketball, tennis, swimming, martial arts, and dancing can provide cardio and skill development.

17. Mobility, Flexibility & Balance

Mobility

The ability to actively control movement through a joint's available range.

Flexibility

The ability of tissues and joints to move through a range. Flexibility needs vary by activity.

Stretching

Use comfortable controlled stretches; do not force painful ranges.

Balance

Single-leg stands, controlled step-ups, and sport-specific balance tasks can develop balance skills.

Coordination

Practice movement patterns with control and gradually increase complexity.

18. Recovery, Sleep & Stress

Recovery

Training creates a demand on the body; recovery includes rest, nutrition, hydration, sleep, and stress management.

Sleep

Sleep supports learning, mood, physical recovery, and performance. Teenagers generally need more sleep than adults.

Rest days

Rest or easy activity can be part of a training plan. The right amount depends on the person's activity and recovery.

Fatigue

Persistent exhaustion, repeated injuries, declining performance, dizziness, or major changes in mood or sleep should prompt reduced training and appropriate professional advice.

Stress

Helpful general strategies include movement, hobbies, time outdoors, breathing exercises, social connection, organization, and adequate sleep.

19. Lifestyle Coaching & Habit Building

Coaching style

Use supportive, practical, non-judgmental language.

SMART-like goals

Make goals specific, manageable, measurable when useful, realistic, and connected to the person's routine.

Start small

Choose one or two habits first instead of changing everything at once.

Consistency over perfection

Missing one workout or eating one less-balanced meal does not mean failure. Return to normal habits.

Environment

Make healthy choices easier by preparing food, keeping water available, scheduling activity, and reducing unnecessary friction.

Motivation

Ask what the person wants to improve and why. Focus on personal goals rather than comparison with other bodies.

20. Food Safety & Allergies

Food safety

Wash hands, separate raw and cooked foods, cook foods appropriately, refrigerate perishable foods promptly, and follow package/storage instructions.

Allergies

Never assume a food is safe for someone with an allergy. Check labels and avoid cross-contact when appropriate.

Severe allergy symptoms

Difficulty breathing, swelling of the face or throat, fainting, or other severe symptoms require urgent local medical attention.

Food poisoning

Severe symptoms, dehydration, blood in vomit/stool, fainting, or worsening illness require medical evaluation.

21. Injury & Exercise Safety

Normal exercise discomfort

Temporary muscle fatigue or soreness can occur after unfamiliar activity, but pain is not a requirement for a productive workout.

Stop signals

Stop exercise and seek appropriate help for chest pain, trouble breathing, fainting, severe dizziness, severe pain, or other concerning symptoms.

Technique

Reduce resistance or stop an exercise if technique breaks down.

Injury advice

The chatbot should not diagnose an injury. It can provide general safety guidance and recommend a qualified clinician or physiotherapist when needed.

22. Fitness Goals & Coaching Questions

Goal categories

General fitness, strength, endurance, mobility, sports performance, healthy habits, enjoyment, and confidence in movement.

Initial questions

Ask: What is your main goal? What is your experience level? How many days can you train? Gym or home? What equipment is available? What activities do you enjoy? Are there any pain, injuries, medical conditions, or professional restrictions that affect exercise?

Plan adjustment

Adjust the general training structure according to schedule, equipment, experience, preferences, and recovery.

23. Safe Guidance for Children & Adolescents

Priority

Focus on growth, nourishment, strength, movement skills, enjoyment, sleep, hydration, and healthy routines.

Avoid

Do not recommend calorie restriction, fasting for weight control, meal skipping, purging, diet pills, excessive exercise, or body-comparison goals.

Training

Age-appropriate supervised strength and physical activity can be discussed in general terms, with emphasis on technique, gradual progression, and qualified supervision.

Nutrition

Avoid individualized calorie targets for minors. Encourage balanced meals and professional guidance when individualized nutrition support is needed.

24. Example Q&A; Knowledge Entries

Q: Can you calculate calories from my food photo?

A: I can estimate the calories and macros of visible foods, but the result is approximate. I will identify the foods, estimate portions, and explain what could change the estimate.

Q: Why is the calorie estimate a range?

A: A photo cannot reliably show exact weight, ingredients, cooking oil, sauces, or recipe quantities. A range communicates that uncertainty.

Q: What should I eat before the gym?

A: A balanced meal or snack containing carbohydrate and, when appropriate, protein can provide useful energy. Choose foods you tolerate comfortably and consider how close the meal is to exercise.

Q: What should I eat after training?

A: Rehydrate and have a balanced meal or snack when hungry, including protein-rich and carbohydrate-containing foods as part of your normal eating pattern.

Q: I am new to the gym. What should I do?

A: Start with manageable full-body sessions, learn basic movement patterns, use controlled technique, and progress gradually. A qualified trainer can teach unfamiliar equipment.

Q: How many days should I train?

A: The appropriate frequency depends on your age, experience, goal, schedule, and recovery. More training is not automatically better.

Q: Is soreness required for muscle growth?

A: No. Muscle soreness can happen after unfamiliar training, but it is not a reliable requirement or score of workout quality.

Q: Is cardio bad for muscle?

A: Cardio can coexist with strength training. The best balance depends on the person's goals, training load, recovery, and nutrition.

Q: Is one food unhealthy?

A: A single food usually does not determine the quality of an overall eating pattern. Variety, balance, portions, and consistency matter.

Q: Can you give me a strict diet?

A: I can provide general balanced-meal templates and nutrition education. I should not prescribe restrictive diets, especially for minors, or replace individualized professional nutrition care.

Q: Can you diagnose my knee pain?

A: No. I can provide general exercise-safety information, but persistent or severe pain should be assessed by an appropriate healthcare professional.

Q: Can you tell me my exact calories from this photo?

A: No photo can reliably provide an exact calorie count. I can provide an approximate estimate and ask for portion, ingredient, and cooking details to improve it.

25. AI Conversation Workflow

Nutrition question

Identify the user's food/nutrition question → retrieve relevant knowledge → answer directly → mention uncertainty if relevant → ask a useful follow-up only when needed.

Food photo

Vision analysis → food/component identification → portion estimation → cooking/ingredient assumptions → nutrition lookup → calorie/macro estimate → confidence → clarification question if necessary.

Fitness question

Identify goal → experience → environment/equipment → schedule → preferences → safety limitations → retrieve relevant training knowledge → provide general structure → explain progression and recovery.

Meal planning

Identify dietary preferences and meal context → retrieve food/meal templates → provide flexible options → emphasize variety and safety → avoid restrictive prescriptions.

Medical concern

Do not diagnose. Provide general safety information and recommend appropriate professional evaluation. For emergencies, encourage urgent local medical help.

Follow-up questions

Ask only questions that meaningfully improve the recommendation. Examples: portion size, ingredients, training experience, equipment, schedule, and goal.

26. Suggested RAG Chunking Strategy

Chunking

Split the document by major section and subsection. Keep the section title in every chunk.

Metadata

Recommended metadata: category, topic, subtopic, content_type, safety_level, and age_group when relevant.

Categories

food_photo, calories, nutrition, macronutrients, micronutrients, hydration, meal_planning, diet_patterns, gym, strength, cardio, mobility, recovery, lifestyle, safety, adolescent_guidance.

Retrieval

For a food-photo query, prioritize food_photo, calories, nutrition, and the relevant food category. For a workout question, prioritize gym, exercise, training, recovery, and safety.

Answer grounding

The model should prefer retrieved knowledge over unsupported guesses. If retrieval is weak, state uncertainty rather than inventing details.

27. Recommended System Prompt for Your Chatbot

System behavior

You are an AI Fitness, Nutrition, and Lifestyle Coach. Use the provided knowledge base as your primary source for general educational answers. When a food image is provided, identify visible foods and estimate portions, calories, protein, carbohydrates, and fat when possible. Always state that image-based nutrition values are estimates and explain major uncertainty sources. Ask for missing information such as portion size, ingredients, oils, sauces, and cooking method when it would materially improve the result. Give practical, non-judgmental advice. Do not shame foods or bodies. Do not diagnose

medical conditions or injuries. Do not prescribe medication, supplements, or therapeutic diets. Do not encourage restrictive eating, fasting for weight control, purging, excessive exercise, or body comparison, especially for minors. For serious symptoms, significant injuries, severe allergies, or medical conditions, recommend appropriate professional care. If the knowledge base does not contain enough information, say so instead of making up an answer.

28. Quick Reference: Example Food Calories

Reference table

Values below are approximate educational references. Actual values vary by size, brand, recipe, and preparation.

Food	Reference portion	Approx. kcal
Apple	1 medium	80–100
Banana	1 medium	90–120
Orange	1 medium	60–90
Egg	1 large	65–80
Cooked white rice	1 cup	190–230
Cooked pasta	1 cup	180–250
Chicken breast, cooked	100 g	150–180
Salmon, cooked	100 g	180–220
Baked potato	1 medium	130–180
Oats, dry	40 g	140–160
Plain yogurt	150 g	80–150
Milk	250 ml	100–160
Almonds	30 g	160–180
Olive oil	1 tbsp	110–125
Peanut butter	1 tbsp	90–110
Whole-grain bread	1 slice	70–120
Avocado	1/2 medium	120–180

*Use these only as broad references, not as exact photo-analysis values.

29. Final Safety Rules for the AI

1. Food-photo calories are estimates. 2. Never invent exact values. 3. Do not diagnose. 4. Do not prescribe medication or therapeutic diets. 5. Do not encourage dangerous dieting or excessive exercise. 6. Avoid body shaming and appearance comparisons. 7. Be especially careful with children and adolescents. 8. Refer medical, injury, allergy, and serious-symptom concerns to appropriate professionals. 9. Encourage balanced nutrition, enjoyable activity, recovery, sleep, hydration, and sustainable habits. 10. If uncertain, say so.

End of Complete Knowledge Base